

Bui Cong Vinh

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Summary

Final-year Computer Science student passionate about large-scale system architecture and production-grade engineering. Proven hands-on experience in a real-world codebase (**TypeScript**, **C#/.NET**) serving actual users during an internship at Laztar. Proficient across **JavaScript/TypeScript**, and **C#**, with daily integration of **AI-assisted tools** (Claude Code, Cursor) to accelerate feature delivery while retaining full code review ownership. Fast learner and confident communicating and collaborating in professional English settings.

Education

International University, Vietnam National University – Ho Chi Minh City

Sept 2022 – Mar 2027

Bachelor of Science in Computer Science – GPA 3.29/4.0

Experience

Laztar

June 2026 – August 2026

Software Engineer Intern, Ho Chi Minh City, Vietnam

Digital transformation and SaaS solutions company based in Ho Chi Minh City, working within a 10–12-person engineering team.

Tech: TypeScript, Next.js, React Native / Expo, NestJS, Prisma, C#/.NET 8, ABP Framework, PostgreSQL, Redis

- Fixed a bug in the **composite-resource booking logic** where creating one resource incorrectly scheduled related resources elsewhere in the hierarchy, caught and resolved before reaching production.
- Diagnosed and fixed an urgent **concurrency bug** where simultaneous bookings returned a generic server error; replaced a lock-based bottleneck with a **database constraint check** to safely handle race conditions.
- Built **client-side reconciliation** for a network-drop edge case where a booking succeeded on the backend but the response never reached the client, avoiding an unnecessary database schema change per team guidance.
- Standardized **error handling** across **2–3** core booking flows, replacing generic failure messages with actionable errors and resolving a state inconsistency where failed requests left slots locked.
- Utilized **Claude Code** and **Cursor** daily to boost routine development speed by **~30%**, consistently diagnosing and shipping tickets within a single working session while maintaining strict **code review ownership**.

NAB Innovation Centre Vietnam

Nov 2025 – Apr 2026

Student Ambassador, NAB Talent Connect Program – Part-time, Remote

- Represented the company on campus for six months, engaging students on engineering culture and early-career opportunities. (Certificate).

Projects

Land Registry on Blockchain – Privacy-Preserving Title Verification ([GitHub Repository Link](#))

Solo Project

Tech: TypeScript, Circom/Groth16, Solidity, NestJS, Next.js, PostgreSQL/Prisma, Hardhat

- Solved citizen privacy and fraud risks in land management by architecting an off-chain **Sparse Merkle Tree (SMT)** with **Groth16 zk-SNARKs**, allowing owners to prove ownership, clean mortgage status, and transfer rights without disclosing identity or underlying records.
- Designed 3 Circom circuits (up to **28.2K** R1CS constraints, depth-24 Merkle tree); achieved **660–1,133 ms** proving time, **14–18 ms** verification, **~722 B** proofs.
- Cut on-chain costs by decoupling storage from Ethereum Sepolia, keeping fees low and predictable as the registry grows: a flat **~115K gas** per root update and **~250K–267K gas** per proof verification.
- Kept the system fast at scale: proof lookups stayed at **~10 ms** p50 even at **4.5M** property records, and caching cut API p99 latency **96.5%** (1,003 ms to 35 ms) for snappier responses under load.
- Self-audited and closed 3 vulnerabilities (proof replay, an exposed public endpoint, a transfer-window ownership race); validated across **22** Sepolia rounds with **740** automated tests.

Awards & Certifications

- **Second Runner-up – AI Thuc Chien 2025: IUAI** (December 2025)
- **Runner-up – IT Hackathon 2025: Solana Pragmatic** (May 2025)
- **Academic Scholarship for Semester II, Academic Year 2025 – 2026**
- **IELTS Academic 6.5** (08/2026, British Council)